Ch 9.1: Maximum Margin Classifier

Lecture 27 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Mon, Nov 13, 2023

Announcements

Last time:

- Ch 8: Random Forests

This lecture:

- Maximal Margin Classifier
- No jupyter notebook for this class

Announcements:

- HW #7....
- No class (virtual OH only) Weds, Nov 22

Status	Lec #	Date			Reading	Homeworks
		Mon	Oct 23	No class - Fall break		
		Wed	Oct 25	**Midterm #2**		
Done	20	Fri	Oct 27	Dimension Reduction	6.3	
Done	21	Mon	Oct 30	More dimension reduction; High dimensions	6.4	
Done	22	Wed	Nov 1	Polynomial & Step Functions	7.1,7.2	
Pushed	23	Fri	Nov 3	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
	24	Mon	Nov 6	Regression Splines	7.4	~~HW #6 Due~~
	25	Wed	Nov 8	Decision Trees	8.1	HW #6 Due
	26	Fri	Nov 10	Random Forests	8.2.1, 8.2.2	
	27	Mon	Nov 13	Maximal Margin Classifier	9.1	
	28	Wed	Nov 15	SVC	9.2	
	29	Fri	Nov 17	SVM	9.3, 9.4	
	30	Mon	Nov 20	Single layer NN	10.1	
	31	Wed	Nov 22	Virtual: Project office hours		
		Fri	Nov 24	No class - Thanksgiving		
		Mon	Nov 27	**Review**		

Section 1

Maximal Margin Classifier

The goal

also divides the classes

Classification

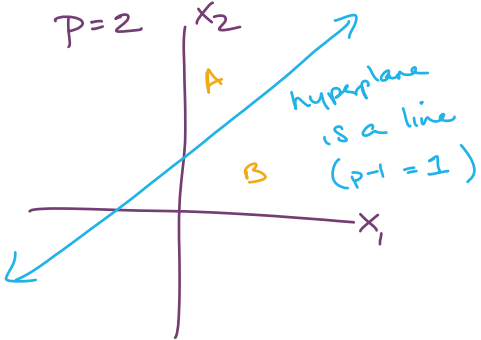
e.g. circles vs stars given (x_1, x_2)

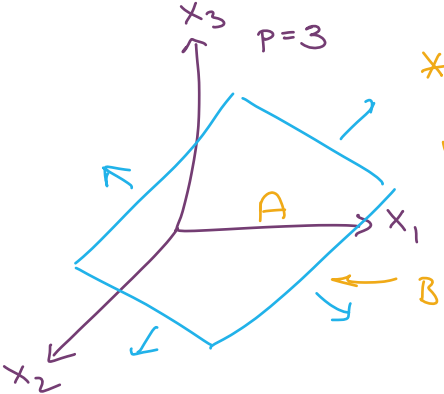
Goal: find line in the middle that separates the two classes as far as possible from given labeled points

Today: predict $y \in \{-1, 1\}$

What is a hyperplane?

↳ In p-dimension space, a hyperplane is a flat affine subspace of dim. p-1





* every hyperplane has two sides

Mathematical definition of a hyperplane

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0$$

$\rightarrow$ Fix $\beta_0, \ldots, \beta_p$

Hyperplane is all $(X_1, \ldots, X_p)$ that satisfy this equation

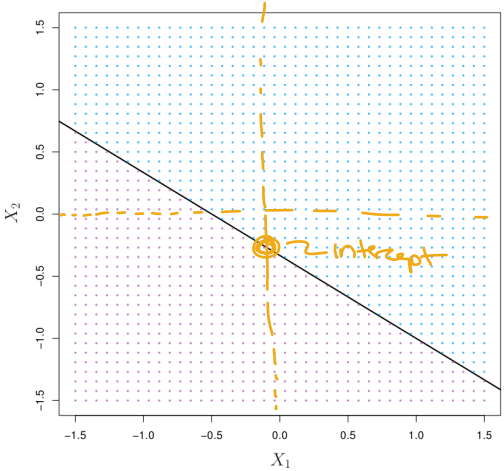
e.g. $p = 2$

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$$

$$\beta_2 X_2 = -\beta_0 - \beta_1 X_1$$

$$X_2 = \underbrace{-\frac{\beta_0}{\beta_2}}_{\text{int.}} - \underbrace{\frac{\beta_1}{\beta_2}}_{\text{slope}} X_1$$

Hyperplane for $p = 2$



$$1 + 2X_1 + 3X_2 = 0$$

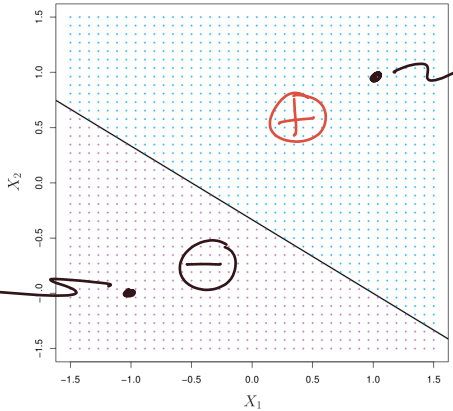
$$1 + 2x_1 + 3x_2 = 0$$

$$x_2 = \frac{-1}{3} - \frac{2}{3}x_1$$

There are two sides to every hyperplane

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p < 0 \qquad \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p > 0$$

$1 + 2x_1 + 3x_2 = 0$



point $(1,1)$

$1 + 2(1) + 3(1) = 6 > 0$

$(-1,-1)$

$1 + 2(-1) + 3(-1)$

$= -4 < 0$

Cheap trick

$-1 - 2x_1 - 3x_2 = 0$

will switch which side has which label

Classification Setup

Data matrix:

$$X = \begin{pmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{pmatrix}_{n \times p}$$

Observations in one of two classes, $y_i \in \{-1, 1\}$

$$Y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$$x_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{1p} \end{pmatrix}, \cdots, x_n = \begin{pmatrix} x_{n1} \\ \vdots \\ x_{np} \end{pmatrix}$$

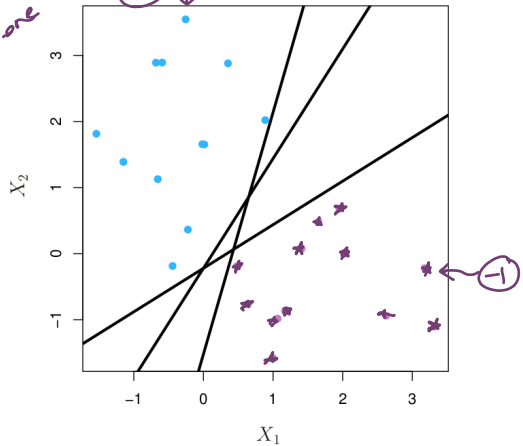
Separate out a test observation

$$x^* = (x_1^* \ \cdots \ x_p^*)^T$$

Separating Hyperplane

Strict requirement: No one on the wrong side of the line

$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0 \text{ if } y_i = 1$$
$$\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0 \text{ if } y_i = -1$$



A hyperplane (choice of $\beta_0, \ldots, \beta_p$) is a separating hyperplane if for every training data point if the sign (pos or neg) of $\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$ is the same as its label ($y_i \in \{-1, 1\}$)

Another way to say it

A $\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} > 0$ if $y_i = 1$

B $\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} < 0$ if $y_i = -1$

all data points

For all i:

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) > 0$$

correct sep hyperplanes

A + +

B − −

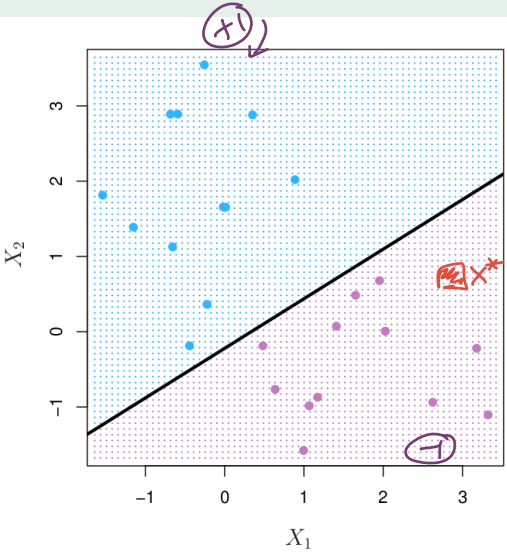
Separating hyperplane becomes a classifier

If you have a separating hyperplane:

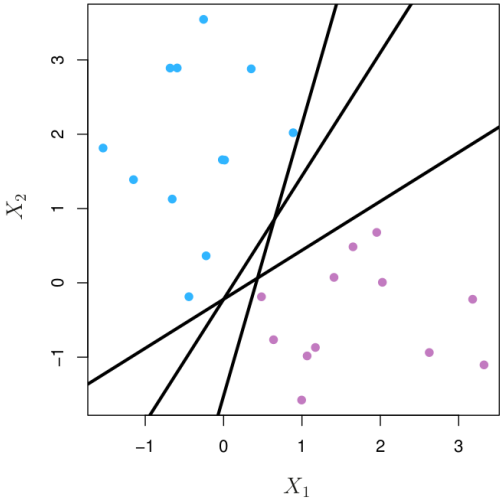
- Check

$$f(x^*) = \beta_0 + \beta_1 x_1^* + \beta_2 x_2^* + \cdots + \beta_p x_p^*$$

- If positive, assign $\hat{y} = 1$
- If negative, assign $\hat{y} = -1$



How do we pick?



Lots of separating hyperplanes

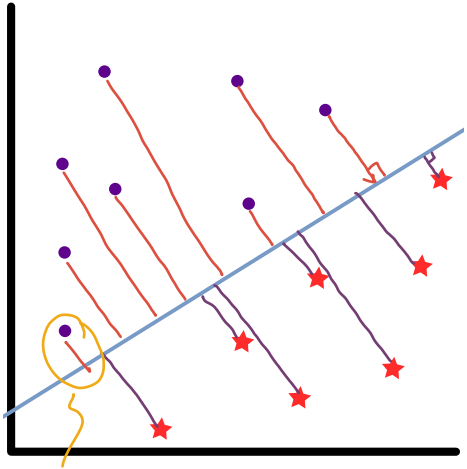
Which is best?

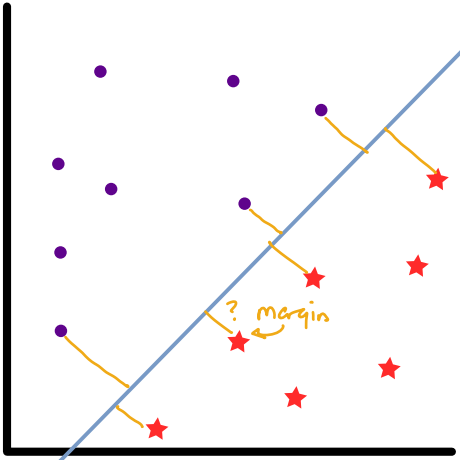
Compute the distance from each training obs to the hyperplane

min distance → "margin"

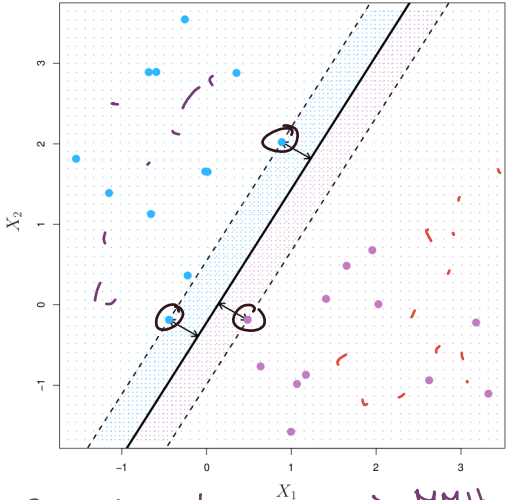
make this as large as possible

Distance from an observation to a hyperplane





Maximal margin classifier



MMH

Maximal margin hyperplane:

the separating hyperplane with largest minimum dist to the training observations

margin

Maximal margin classifier:

turn this hyperplane into classifier

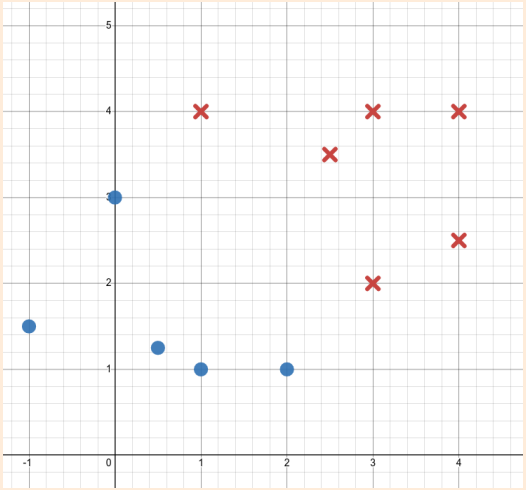
Support vectors:

the observations (training data points) that are closest to the hyperplane

If support vectors move ⇒ MMH move

If any other data moves ⇒ no change

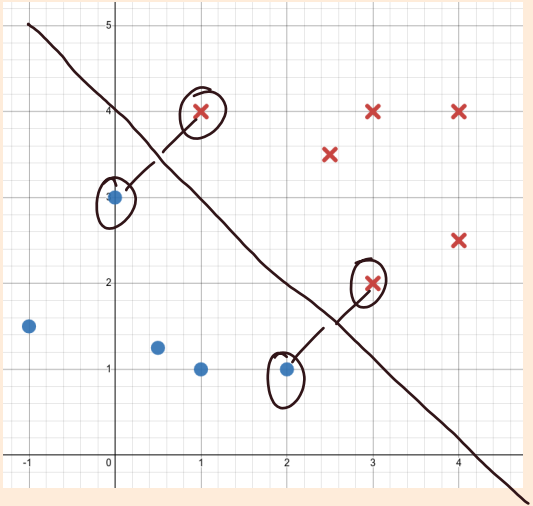
Example



- Sketch the maximal margin hyperplane.
- What is the equation of this line in the form $\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0$?
- Circle the support vectors. What is their distance from the line?

desmos.com/calculator/lqms253gfq

Extra work space



$$x_2 = 4 + (-1)\,x_1$$

$$x_1 + x_2 - 4 = 0$$

distance $\frac{\sqrt{2}}{2}$ = margin

$$x_1 + x_2 - 4 = 0$$

$$2x_1 + 2x_2 - 8 = 0$$

desmos.com/calculator/lqms253gfq

Mathematical Formulation

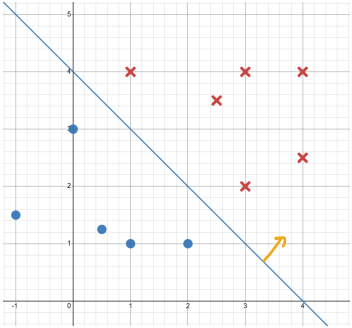
$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \quad M > 0$$

forces sum of squares to be 1

$$\text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$\underbrace{y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip})}_{\geq 0 \text{ if the classifier is separating}} \geq M \quad \forall\, i = 1, \ldots, n$$

First constraint



$$\sum \beta_i^2 = 1$$

$$-4 + x_1 + x_2 = 0$$

divide by $\sqrt{\sum_{i=1}^{2} \beta_i^2}$

$$= \sqrt{1+1}$$

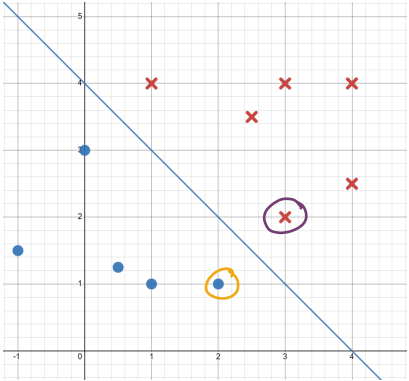
$$= \sqrt{2}$$

$$\boxed{\frac{-4}{\sqrt{2}} + \underbrace{\frac{1}{\sqrt{2}} x_1 + \frac{1}{\sqrt{2}} x_2}_{} = 0}$$

$$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2$$

$$\frac{1+1}{2} = 1$$

Second constraint



- Blue circles: $y_i = -1$
- Red Xs: $y_i = 1$
- $-2\sqrt{2} + \frac{\sqrt{2}}{2}X_1 + \frac{\sqrt{2}}{2}X_1 = 0$ ← pretend I got this last page

the dist to the line

maximized at the margin

$$\underbrace{y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})}_{} \geq M$$

eg) $(3,2)$

$$(1)\left(-2\sqrt{2} + \frac{\sqrt{2}}{2}(3) + \frac{\sqrt{2}}{2}(2)\right)$$

$$-2\sqrt{2} + \frac{3}{2}\sqrt{2} + \sqrt{2}$$

$$(-2 + 1.5 + 1)\sqrt{2} = 0.5\sqrt{2} = \frac{\sqrt{2}}{2}$$

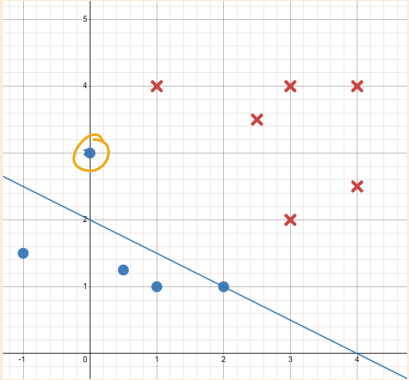
eg) $(2,1)$

$$(-1)\left(-2\sqrt{2} + \frac{\sqrt{2}}{2}(2) + \frac{\sqrt{2}}{2}(1)\right)$$

$$= (-1)\left(-2 + 1 + \tfrac{1}{2}\right)\sqrt{2}$$

$$= (-1)\left(-\tfrac{1}{2}\right) = \frac{\sqrt{2}}{2}$$

An example with a bad choice of hyperplane



- Blue circles: $y_i = -1$
- Red Xs: $y_i = 1$
- $-\frac{4}{\sqrt{5}} + \frac{1}{\sqrt{5}}X_1 + \frac{2}{\sqrt{5}}X_1 = 0$

What is $y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2})$ for the point $x_i = (0, 3)$?

$$(-1)\left(-\frac{4}{\sqrt{5}} + \frac{1}{\sqrt{5}} \cdot 0 + \frac{2}{\sqrt{5}} \cdot 3\right)$$

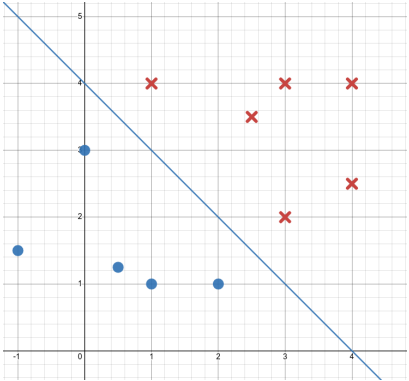
$$(-1)\left(\frac{2}{\sqrt{5}}\right)$$

This is negative!

requirement.... this $\geq M > 0$

No possible positive M makes this true!

Second constraint extra space



- Blue circles: $y_i = -1$
- Red Xs: $y_i = 1$
- $-2\sqrt{2} + \frac{\sqrt{2}}{2}X_1 + \frac{\sqrt{2}}{2}X_1 = 0$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \geq M$$

ensures

* all points on the correct side of the line
* and the margin is as large as possible

Mathematical Formulation

$$\underset{\beta_0,\beta_1,\ldots,\beta_p,M}{\text{maximize}} \; M$$

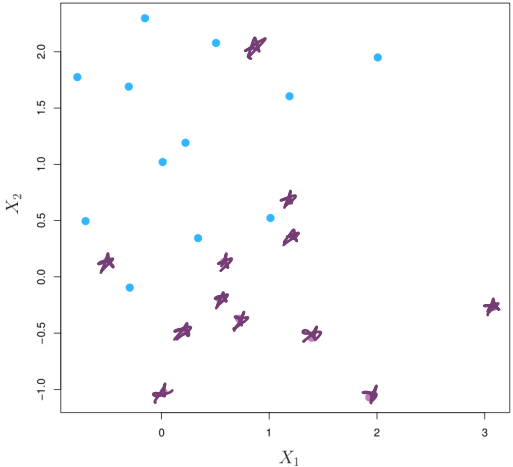
$$\text{subject to } \sum_{j=1}^{p} \beta_j^2 = 1,$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}) \geq M \;\; \forall\, i = 1, \ldots, n$$

Section 2

Issues with Maximal Margin Classifier

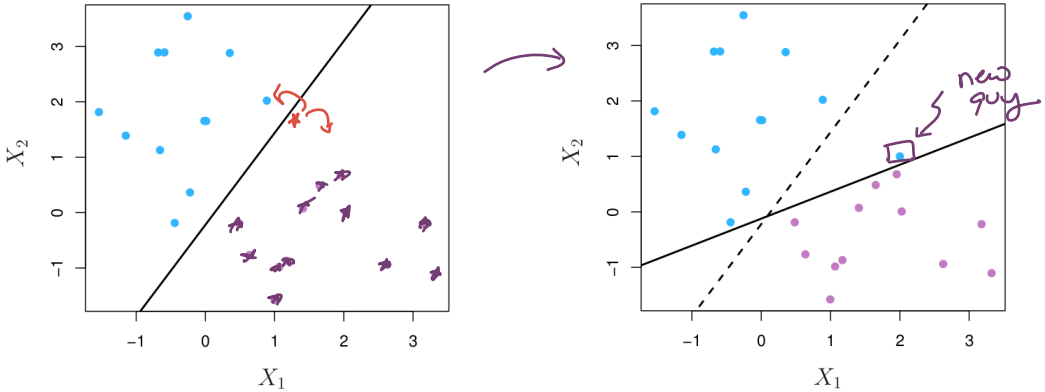
But what if....



There is no separating hyperplane!

↳ no solution to the optimization problem for any $M > 0$

Sensitivity to new points



Even if there is one it might not be good and very sensitive to new pts

Next time

Status	Lec #	Date			Reading	Homeworks
		Mon	Oct 23	No class - Fall break		
		Wed	Oct 25	**Midterm #2**		
Done	20	Fri	Oct 27	Dimension Reduction	6.3	
Done	21	Mon	Oct 30	More dimension reduction; High dimensions	6.4	
Done	22	Wed	Nov 1	Polynomial & Step Functions	7.1,7.2	
Pushed	23	Fri	Nov 3	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
	24	Mon	Nov 6	Regression Splines	7.4	~~HW #6 Due~~
	25	Wed	Nov 8	Decision Trees	8.1	HW #6 Due
	26	Fri	Nov 10	Random Forests	8.2.1, 8.2.2	
	27	Mon	Nov 13	Maximal Margin Classifier	9.1	
	28	Wed	Nov 15	SVC	9.2	
	29	Fri	Nov 17	SVM	9.3, 9.4	
	30	Mon	Nov 20	Single layer NN	10.1	
	31	Wed	Nov 22	Virtual: Project office hours		
		Fri	Nov 24	No class - Thanksgiving		
		Mon	Nov 27	**Review**		
		Wed	Nov 29	**Midterm #3**		
	32	Fri	Dec 1	Multi Layer NN	10.2	
	33	Mon	Dec 4	CNN	10.3	
	34	Wed	Dec 6	Unsupervised Learning & Clustering	12.1, 12.4	
	35	Fri	Dec 8	Virtual: Project office hours		Project due